

# Closed forms for the constants in the odd-weight identities between dihedral modular graph functions

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**Abstract** Dihedral modular graph functions  $C(a, b, c)$  of weight  $w = a + b + c$  are the coefficients of the low-energy expansion of the one-loop closed-string amplitude. For each odd weight there is a combination  $X_w$  whose Laplacian is a multiple of  $E(w)$ . We determine the constants in  $X_w = f_w E(w) + g_w \zeta(w)$ . Theorem 1 gives  $f_w = 3((w-1)/2)!/w$  and  $g_w = 6|B_{w-1}|/((w-1)/2)!$  for the odd weights  $3 \leq w \leq 61$  examined. The proof combines exact rational Laplace algebra, a harmonicity argument and the Laurent constant term evaluated in rational arithmetic. It rests on two assumptions that were not machine-checked: a published theorem on Laurent polynomials and a standard lemma. Theorem 2 shows exactly, for the weights  $3 \leq w \leq 25$  examined, that the space of combinations with  $\Delta X \in \mathbb{Q} E(w)$  has dimension 1 for odd  $w$  and 0 for even  $w$ . Independent numerical confirmation at 32 digits covers  $w = 3, 5, 7, 9, 11, 13, 15, 17$ , with  $w = 15$  a prediction and  $w = 17$  a preregistered blind test. The full weight-11 basis has exactly one relation numerically. A proof beyond the examined range is stated as a conjecture and remains open.

**Keywords** Modular graph functions · Laplace eigenvalue equations · Eisenstein series · Bernoulli numbers · Genus-one string amplitudes · Computer-assisted proof

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## 1 Introduction

Two-loop modular graph functions  $C(a, b, c)$  satisfy Laplace eigenvalue equations with inhomogeneous terms that are polynomial in non-holomorphic Eisenstein series [1]. D'Hoker, Green and Vanhove (DGV) obtained, for each odd weight, a combination  $X_w$  that is annihilated by the Laplacian after subtracting a multiple of  $E(w)$ . The remaining additive constant of integration is fixed by the asymptotics at the cusp [1]. DGV give  $f_w$  for  $w \leq 9$  and  $g_w$  for  $w = 3$  and  $w = 5$ , while  $g_7$  and  $g_9$  remain undetermined there. The value of  $g_7$  appears explicitly in later work [2].

**Open question.** What are the exact closed forms of  $f_w$  and  $g_w$  for general odd  $w$ ?

**Why it is hard.** The constant is the  $\tau_2^0$  term of a Laurent polynomial. That polynomial is known for each single  $C_{u,v,w}$ , but evaluating it for the DGV combination requires exact bookkeeping of growing coefficient

lists. Numerical relation searches are limited by the coefficient height; for the full weight-11 basis the originally preregistered completeness criterion was failed.

**Contribution.** We evaluate the Laurent constant of  $X_w$  exactly for the odd weights  $3 \leq w \leq 61$  examined, obtain closed forms, and confirm them independently by numerics.

### Summary of results.

- Theorem 1 shows  $X_w = f_w E(w) + g_w \zeta(w)$  with  $f_w = 3((w-1)/2)!/w$  and  $g_w = 6|B_{w-1}|/((w-1)/2)!$  for the odd weights  $3 \leq w \leq 61$  examined (exact Laplace algebra plus Laurent constant in rational arithmetic).
- Theorem 2 shows that the space  $\{X : \Delta X \in \mathbb{Q} E(w)\}$  has dimension 1 for odd  $w$  and 0 for even  $w$ , for the weights  $3 \leq w \leq 25$  examined.
- Numerical results, independent of the proof, confirm the constants at  $w = 3, 5, 7, 9, 11, 13, 15, 17$  and show that the full weight-11 basis has exactly one relation.

## 2 Setting

Let  $\tau = \tau_1 + i\tau_2$ ,  $q = e^{2\pi i \tau}$  and  $p = m\tau + n$ . The lattice sums use the normalisation  $(\tau_2/(\pi|p|^2))^a$ :

Here  $(a_1, a_2, a_3) = (a, b, c)$  and  $w = a + b + c$ .  $\zeta(k)$  is the Riemann zeta value and  $B_n$  are Bernoulli numbers. The Laplacian is  $\Delta = \tau_2^2(\partial_{\tau_1}^2 + \partial_{\tau_2}^2)$ .  $E(w)$  is an eigenfunction,  $\Delta E(w) = w(w-1)E(w)$ . For odd  $w$ ,  $X_w$  is the DGV combination (eq. 3.57), normalised so that  $\Delta X_w = w(w-1)f_w E(w)$ .

Numerical parameters of the verifier:

- four random points, seed 4711, with  $|\tau_1| \leq 1/2$  and  $1 \leq \tau_2 \leq 2.5$ ;
- working precision of 32 digits;
- tolerance 1e-24 for relative residuals, and 1e-14 when a Laplacian is involved;
- fourth-order finite-difference Laplacian with step  $h = 1e-6$ .

## 3 Closed forms for odd weight

**Theorem 1** (Closed forms,  $w \leq 61$ ). *For the odd weights  $3 \leq w \leq 61$  examined,*

$$X_w = f_w E(w) + g_w \zeta(w), \quad f_w = \frac{3((w-1)/2)!}{w}, \quad g_w = \frac{6|B_{w-1}|}{((w-1)/2)!}$$

**Table 1** Constants  $f_w$  and  $g_w$  in  $X_w = f_w E(w) + g_w \zeta(w)$  (DGV normalisation of  $X_w$ ), exact for odd  $3 \leq w \leq 25$ .

| $w$ | $f_w$        | $g_w$                  |
|-----|--------------|------------------------|
| 3   | 1            | 1                      |
| 5   | 6/5          | 1/10                   |
| 7   | 18/7         | 1/42                   |
| 9   | 8            | 1/120                  |
| 11  | 360/11       | 1/264                  |
| 13  | 2160/13      | 691/327600             |
| 15  | 1008         | 1/720                  |
| 17  | 120960/17    | 3617/3427200           |
| 19  | 1088640/19   | 43867/48263040         |
| 21  | 518400       | 174611/199584000       |
| 23  | 119750400/23 | 77683/83462400         |
| 25  | 57480192     | 236364091/217945728000 |

*Proof.* The proof has three steps and a fourth that supplies the constant.

1. Exact rational computation in the algebraic Laplace representation gives  $\Delta X_w = w(w-1)f_w E(w)$  with no other terms. Since  $\Delta E(w) = w(w-1)E(w)$ , it follows that  $\Delta(X_w - f_w E(w)) = 0$ .
2.  $X_w - f_w E(w)$  is  $SL(2, \mathbb{Z})$ -invariant and of polynomial growth at the cusp. By the standard argument used by DGV it is therefore constant.
3. The constant is the  $\tau_2^0$  term of the Laurent polynomial of  $X_w$ . It was computed in rational arithmetic from Proposition 2.1 and Theorem 5.1 of D'Hoker-Kaidi, with the transcription checked against their eq. 5.19. All Laurent terms not involving products of zeta values agree with those of  $f_w E(w) + g_w \zeta(w)$  (consistency check). The constant equals  $g_w \zeta(w)$  with  $g_w = 6|B_{w-1}|/((w-1)/2)!$  for the odd weights  $3 \leq w \leq 61$  examined.

Assumptions not checked by code are the correctness of Theorem 5.1 of D'Hoker-Kaidi and the standard lemma in step 2. Theorem 5.1 is proved there in its Appendix A; its conjectural part concerns only the coefficient of  $\tau_2^{2-w}$  and is not used. Examples are  $g_3 = 1$ ,  $g_5 = 1/10$ ,  $g_7 = 1/42$ ,  $g_9 = 1/120$ .

Novelty status: partly known. Closed forms of  $f_w$  and  $g_w$  for general odd  $w$  and their proof for  $w \geq 9$  are not given there.

#### 4 Uniqueness of the harmonic combination

**Theorem 2** (Dimension of the harmonic space). *For odd  $w$  it is spanned by the DGV combination, with  $\Delta X = w(w-1)f_w E(w)$  and  $f_w = 3((w-1)/2)!/w$ .*

*Certificate.* The statement was verified exactly for the weights  $3 \leq w \leq 25$  examined, with the algebraic Laplace representation of DGV, re-derived and evaluated in rational arithmetic. Novelty status: partly known. The one-dimensionality of the eigenspace for odd weight, its vanishing for even weight and the combination  $X_w$  are known.

**Proposition 1** (Laplace equations at  $w = 9$  and  $w = 11$ ).  $\Delta(9C(4, 4, 1) + 18C(4, 3, 2) + 4C(3, 3, 3)) =$

$$288 E(9) \text{ and } \Delta(2C(5, 5, 1) + 4C(5, 4, 2) + 2C(5, 3, 3) + 3C(4, 4, 3)) = 100 E(11).$$

*Certificate.* Both equations were obtained in the algebraic Laplace representation and checked in exact rational arithmetic. The verifier returns a single value of  $\lambda$  in  $\Delta X = \lambda E(w)$ . Novelty status: the  $w = 9$  equation is already known. For  $w = 11$  the combination is known, but  $f_w$  and  $g_w$  are not given there.

**Conjecture 1** (Beyond the examined range). *The identity  $X_w = f_w E(w) + g_w \zeta(w)$  with the closed forms of Theorem 1 holds beyond the examined range of odd weights.*

Before the proof, this was a conjecture read off from  $w = 3..13$ .

#### 5 Numerical confirmation, independent of the proof

The statements in this section are numerical (32 digits, four verifier-chosen points) combined with the exact vanishing of the leading Laurent coefficient. They do not use the proof of Theorem 1.

**Example 1** (Weight 3).  $C(1, 1, 1) - E(3) - \zeta(3) = 0$  as functions of  $\tau$ .

*Certificate.* The exact leading Laurent coefficient in  $y^3$  vanishes, and at the four verifier points the maximal relative residual is 5.13e-44 against the bound 1e-24. This identity is already known [1].

**Example 2** (Weight 5).  $30 C(2, 2, 1) - 12 E(5) - \zeta(5) = 0$  as functions of  $\tau$ .

*Certificate.* The leading coefficient in  $y^5$  vanishes exactly, and the maximal relative residual is 1.57e-43. This identity is already known [1].

**Example 3** (Weight 7).  $252 C(3, 3, 1) + 252 C(3, 2, 2) - 108 E(7) - \zeta(7) = 0$  as functions of  $\tau$ .

*Certificate.* The leading coefficient in  $y^7$  vanishes exactly, and the maximal relative residual is 1.59e-43. This is  $f_7$  and  $g_7$  in the notation above, already known [2].

**Numerical observation 1** (Weight 9).  $960 E(9) + \zeta(9) - 2160 C(4, 4, 1) - 4320 C(4, 3, 2) - 960 C(3, 3, 3) = 0$  as functions of  $\tau$ .

*Certificate.* The leading coefficient in  $y^9$  vanishes exactly, and the maximal relative residual is 8.33e-44. Novelty status: partly known. DGV give the identity with an undetermined constant  $g_9$ , whose value is not given there.

Here  $E(w)$  is the lattice sum of the Setting and  $\zeta(w)$  the Riemann zeta value at  $w$ .

Each of the four relations has vanishing exact leading Laurent coefficient (in  $y^{11}, y^{13}, y^{15}, y^{17}$ ). The value at  $w = 15$  was a prediction made before its value was read, and  $w = 17$  was a preregistered blind test. Novelty

**Table 2** Numerical checks of the integer-normalised relations at weights 11, 13, 15, 17; the third column is the maximal relative residual at the four verifier points (bound 1e-24).

| $w$ | coefficients of $E(w)$ and $\zeta(w)$ | residual |
|-----|---------------------------------------|----------|
| 11  | $-8640 E(11), -\zeta(11)$             | 1.21e-43 |
| 13  | $-54432000 E(13), -691 \zeta(13)$     | 5.67e-44 |
| 15  | $-725760 E(15), -\zeta(15)$           | 6.46e-44 |
| 17  | $-24385536000 E(17), -3617 \zeta(17)$ | 5.49e-44 |

status: partly known. Existence and combination for each odd  $w$  are known, but  $f_w$  and  $g_w$  for  $w \geq 11$  are not given there.

**Numerical observation 2** (Full weight-11 basis). The rational linear relations among  $C(9, 1, 1)$ ,  $C(8, 2, 1)$ ,  $C(7, 3, 1)$ ,  $C(7, 2, 2)$ ,  $C(6, 4, 1)$ ,  $C(6, 3, 2)$ ,  $C(5, 5, 1)$ ,  $C(5, 4, 2)$ ,  $C(5, 3, 3)$ ,  $C(4, 4, 3)$ ,  $E(11)$ ,  $\zeta(11)$  form a space of dimension exactly 1 (weight 11), spanned by the weight-11 relation of the table.

The threshold was chosen so as to separate the null value 1.11e-43 from the smallest large singular value 3.69e-9. One singular value is below 1e-24 and 11 are above 1e-16. Novelty status: partly known; see the weight-11 entry above.

**Numerical observation 3** (Hardening of the weight-11 identity). An independent direct lattice summation in double precision agrees to at most 1e-12 relative at 8 of 8 points.

*Certificate.* The criterion operationalises the preregistered expectation that the residual decreases with precision; an exact 0 counts as below rounding resolution. The run exposed and fixed an evaluator bug. The Fourier sum for  $E_s$  stopped when a term vanished, which happens at  $\tau_1 = 1/4$  where  $\cos(2\pi N\tau_1) = 0$  for odd  $N$ . Before the fix that point showed a precision-independent residual of 3.5e-10. All listed points were computed with the corrected code, and no earlier test point was affected.

*Remark 1* (Status of the computer-assisted parts). The trusted base consists of the following.

- **Exact rational arithmetic.** It covers the Laplace algebra of Theorems 1 and 2 and the Laurent constants.
- **The 32-digit evaluator.** It covers the numerical observations, with the tolerances of the Setting.

Not machine-checked are Theorem 5.1 of D’Hoker-Kaidi and the harmonic-implies-constant lemma. A third party re-runs all certificates with the commands in the Declarations. Statements marked numerical are not proofs.

## 6 Discussion

**Limitations.** The proof depends on two external ingredients, listed in the status remark. The range  $w \leq 61$  is

a computational bound, not a theorem beyond it. Numerical confirmation certifies values at sampled points together with exact leading coefficients, for the weights  $w = 3, 5, 7, 9, 11, 13, 15, 17$  examined.

### What did not work.

It was reported openly, and criterion 2, preregistered afterwards on new data, passed. - The hardening run found an evaluator bug at  $\tau_1 = 1/4$  (early termination of the Fourier sum when a term vanished). It was fixed, and all listed points use the corrected code.

Further negative results are collected in Appendix A.

### Open questions.

- Is there a proof of  $X_w = f_w E(w) + g_w \zeta(w)$  beyond the examined range, i.e. of Conjecture 1, presumably via a binomial-sum identity for the Laurent constant?
- What are the constants in the non-harmonic Laplace eigenvalue equations for  $C(a, b, c)$ ?
- Do analogous closed forms exist at higher loop order?

## Declarations

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**Code and data availability** Repository <https://github.com/NinjaTurtlesHackathons/Hacknation>, branch `claude/modulformen-paper`, directory `projects/modular`. All certificates are reproduced with `python -m asd.recheck modular`, and the paper is rebuilt with `python -m asd.paper --domain modular`.

**Competing interests** None.

## A Numerical methods and error control

### Certificate types.

- Exact Laplace algebra and Laurent constants:* rational arithmetic, no floating point.
- Function relations:* the leading Laurent coefficient is checked exactly, then numerically at four verifier points (seed 4711, 32 digits, tolerance 1e-24, or 1e-14 with the finite-difference Laplacian of step 1e-6).
- Relation spaces:* completeness is shown by a singular-value gap of the normalised value matrix at independent points (seed 4712).

### Validation of the verifier on known results.

The singular values of the normalised matrices show one value below 1e-24; for  $w = 3$  the matrix is 6x3 with smallest large value 0.731. - Laplace

equations:  $\Delta C(3, 1, 1) - 6C(3, 1, 1) - \frac{86}{5}E(5) + 4E(2)E(3) - \frac{1}{10}\zeta(5) = 0$  with residual  $1.63\text{e-}25$ .  
 $\Delta C(2, 1, 1) - 2C(2, 1, 1) - 9E(4) + E(2)^2 = 0$  with  $2.39\text{e-}25$ .  
 $\Delta C(2, 2, 1) - 8E(5) = 0$  with  $6.99\text{e-}26$ .  
 $\Delta C(2, 2, 1) - 20C(2, 2, 1) + \frac{2}{3}\zeta(5) = 0$  with  $6.74\text{e-}26$ .  
- Holomorphic identities (exact, Sturm certificates):  
-  $\theta(1)^4 = \eta(2)^{20}\eta(1)^{-8}\eta(4)^{-8}$  on  $\Gamma_0(4)$ , with coefficients  $q^0..q^{11}$  vanishing and Sturm bound 1. The weight-1 version  $\theta(1)^2 = \eta(2)^{10}\eta(1)^{-4}\eta(4)^{-4}$  also holds. The relation of eta quotients to theta functions is stated in [3].  
-  $\theta_{E8} = E_4$ ,  $\theta_{E8}^2 = \theta_{D16}$ ,  $\theta_{E8}^2 = E_4^2$  and  $\theta_{E8}^2 = E_8$ .  
-  $1728\Delta = E_4^3 - E_6^2$ , a classical fact stated in [4], and  $j\Delta = E_4^3$ .  
- Eta quotients with trivial character span  $M_4(\Gamma_0(N))$  exactly for  $N \in \{2, 4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 22, 24, 25, 26, 27, 28, 30\}$  and not for  $N \in \{1, 3, 7, 11, 13, 17, 19, 21, 23, 29\}$  (exact, exhaustive,  $N \leq 30$ ).

### Negative results.

- At weight 6, no claim passed for the relation space of  $C(4, 1, 1)$ ,  $C(3, 2, 1)$ ,  $C(2, 2, 2)$  and products of Eisenstein series and zeta values; the 12x9 matrix showed 8 singular values above  $1\text{e-}8$  and none below  $1\text{e-}24$ .
- No claim passed for the eta-span table in the first attempts. The run was aborted, and the presumed cause is precision too low relative to the coefficient height.
- No claim passed in the first attempt at the weight-13 and weight-15 constants. Both relations were later confirmed.

**Laboratory workflow.** The laboratory ran 17 rounds with 37 verified statements and 4 negative results. Every round was preregistered before the experiment. Statements were proposed by agents in the roles *sparsam*, *numeriker*, *skeptiker* and *theoretiker*, and accepted only if the verifier passed them. Adversarial counter-checks (red team) numbered 53 in total; 4 passed without contradicting the statement, 48 did not pass, and 1 was not executable. The verifier rejected falsified variants, such as a doubled  $\zeta(17)$  coefficient or a wrong  $\zeta(5)$  source term.

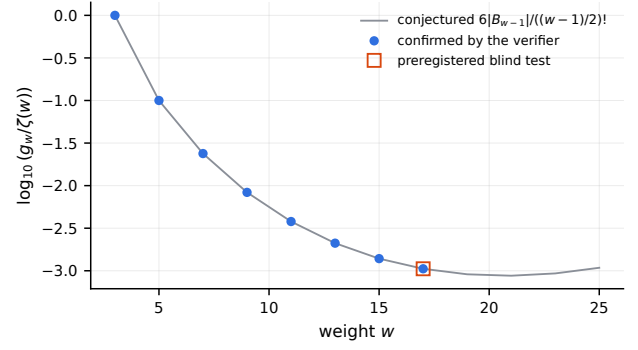
## B Provenance of the statements

The provenance table mapping every statement to its evidence follows (it is generated automatically).

The complete automatic checking protocol of the writing process is stored as `pruefprotokoll.json` in the project directory of the repository.

## References

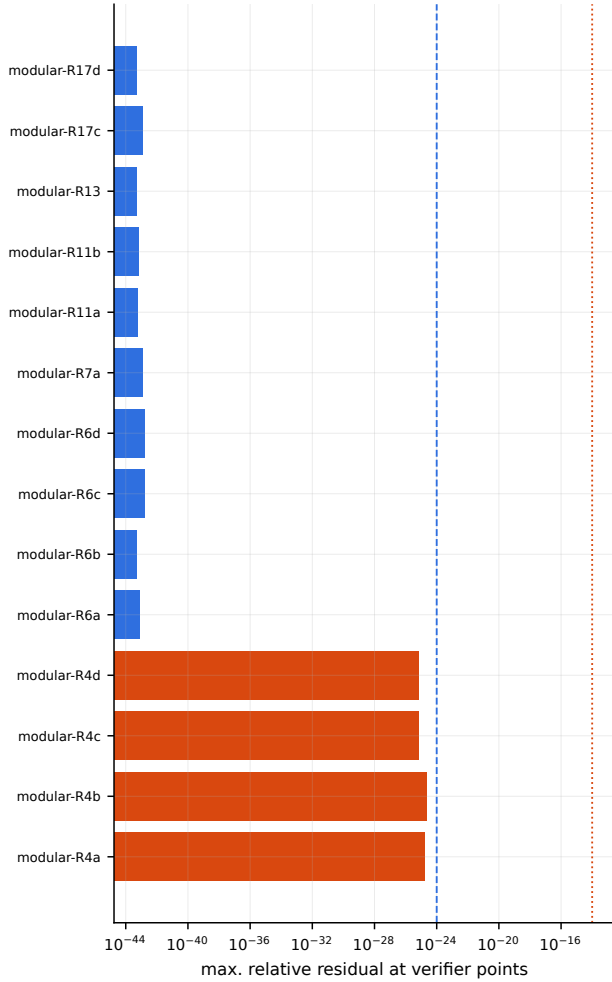
- [1] Eric D'Hoker, Michael B. Green, and Pierre Vanhove. On the modular structure of the genus-one Type II superstring low energy expansion. *arXiv:1502.06698*, 2015.
- [2] Daniele Dorigoni, Axel Kleinschmidt, and Oliver Schlotterer. Poincaré series for modular graph forms at depth two. I. Seeds and Laplace systems. *arXiv:2109.05017*, 2021.



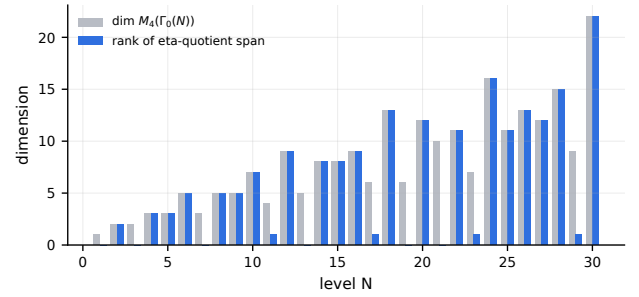
**Fig. 1** Integration constants  $g_w$  of the odd-weight identities in the normalisation of eq. (3.57) of D'Hoker-Green-Vanhove: conjectured closed form (line) and the weights at which the verifier confirmed the identity numerically (dots, 32 digits, 4 verifier points); the square marks the preregistered blind test at  $w = 17$ .

- [3] K. R. Vasuki, Darshan D., and Ravi G. N. On Oliver's relations between infinite series and infinite products. *arXiv:2607.26471v1*, 2026.
- [4] Jay Jorgenson, Lejla Smajlovic, and Holger Then. Certain aspects of holomorphic function theory on some genus zero arithmetic groups. *arXiv:1505.06042v1*, 2015.

| Statement                                                  | Level                       |
|------------------------------------------------------------|-----------------------------|
| Section: Introduction                                      | computed_rigorous, observed |
| Section: Setting                                           | computed_rigorous           |
| Theorem 1 (Closed forms, $w \leq 61$ )                     | computed_rigorous           |
| Section: Closed forms for odd weight                       | computed_rigorous           |
| Theorem 2 (Dimension of the harmonic space)                | computed_rigorous           |
| Section: Uniqueness of the harmonic combination            | computed_rigorous           |
| Proposition 1 (Laplace equations at $w = 9$ and $w = 11$ ) | computed_rigorous           |
| Conjecture 1 (Beyond the examined range)                   | computed_rigorous           |
| Section: Numerical confirmation, independent of the proof  | computed_rigorous, observed |
| Example 1 (Weight 3)                                       | observed                    |
| Example 2 (Weight 5)                                       | observed                    |
| Example 3 (Weight 7)                                       | observed                    |
| Observation 1 (Weight 9)                                   | observed                    |
| Observation 2 (Full weight-11 basis)                       | observed                    |
| Observation 3 (Hardening of the weight-11 identity)        | observed                    |
| Remark 1 (Status of the computer-assisted parts)           | computed_rigorous, observed |
| Section: Discussion                                        | computed_rigorous, observed |
| Section: Declarations                                      | observed                    |
| Section: Appendix A: Numerical methods and error control   | computed_rigorous, observed |



**Fig. 2** Numerical verification margins: largest relative residual of each confirmed relation among modular graph functions at the verifier's own random points (32 digits). Dashed: acceptance threshold  $10^{-24}$ ; dotted: threshold  $10^{-14}$  for relations containing a Laplacian (orange).



**Fig. 3** Exact comparison, for every level  $N \leq 30$ , of  $\dim M_4(\Gamma_0(N))$  (grey) with the rank of the span of all holomorphic eta quotients of weight 4 with trivial character (blue).